

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: MSRIT_Waypoint

Project / Product Name: Agentic Code Intelligence — Semantic Code Retrieval (Theme 1)

Organization / Institution (if any): M. S. Ramaiah Institute of Technology (MSRIT),
Bengaluru

Submission Date: 30 September 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes

Yes — Claude Code (Anthropic, Claude Opus models) was used as a coding assistant throughout. All design decisions, model choices and reported numbers were checked by the team against real benchmark runs.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Yes — discussed pipeline options (hybrid, reranking, query rewriting) with Claude Code; team chose the measured single-stage design.

Code generation or assistance: Yes — Claude Code wrote most of the pipeline, CLI and evaluation code under team direction and review.

UI / UX design: No UI (command-line tool by design).

Content creation: Yes — README, presentation text and this form drafted with Claude Code, edited by the team.

Data analysis: Yes — scripts for the model / query-mode benchmark sweep were AI-assisted; all numbers come from our own runs.

Testing / debugging: Yes — versioning unit tests and bug fixes AI-assisted.

Other: Extraction and summarisation of the official theme guide / FAQ.

4. Feature Origin Classification (Please add/delete based on the number of the feature)

Feature Name: Model selection & benchmark harness (scratch/bench.py, run_dense.py, exp_qmode.py)

Self-Generated/AI-Generated/Both: Both

Description: Tool: Claude Code. Prompt: 'build a fast harness that embeds the APPS corpus once, caches it, and scores NDCG@10/MRR for several code embedding models'. Output: caching harness + sweep over 7 models and 4 query modes. Modification: team chose which models to test, ran every sweep, and picked jina-code-embeddings-0.5b from the measured results.

Feature Name: Query pre-processing (src/preprocess.py)

Self-Generated/AI-Generated/Both: Both

Description: Tool: Claude Code. Prompt: 'APPS queries are LaTeX-heavy problem statements — try cleaning / re-ordering sections before embedding'. Output: clean / noex / iofirst modes. Modification: kept only 'clean' (+0.2 NDCG); re-ordering modes were measured to hurt and were dropped.

Feature Name: Retriever and MTEB evaluation wrapper (src/retriever.py, src/evaluate.py)

Self-Generated/AI-Generated/Both: Both

Description: Tool: Claude Code. Prompt: 'wrap the pipeline as an MTEB search model for AppsRetrieval, CPU-first, and log timings and model size'. Output: CodeRetriever class and MTEB SearchProtocol adapter writing appsretrieval_results.json and a resource profile. Modification: added a --device flag so the bulk benchmark can run on Apple GPU; scores are device-independent.

Feature Name: Versioned index and evolutionary retrieval (src/versioned_index.py, src/chunking.py, src/index.py)

Self-Generated/AI-Generated/Both: Both

Description: Tool: Claude Code. Prompt: 'design an index that re-embeds only changed code when a new version arrives, and search across all versions without returning near-duplicates'. Output: content-addressed (SHA-1) embedding store, per-version manifests, lineage-level ranking, AST chunking. Modification: team reviewed the design and tests; demo on three requests releases.

Feature Name: CLI, Docker, README, presentation

Self-Generated/AI-Generated/Both: Both

Description: Tool: Claude Code. Prompt: 'query CLI for live demos, CPU Docker image, README with reproducible steps, fill in the submission PPT'. Output: src/query.py, Dockerfile, README, PPT text. Modification: reviewed and edited by the team. Demo video: recorded and narrated by the team; Claude Code (ffmpeg + a local Whisper transcription) trimmed filler words and sped up silent stretches to fit the 5-minute limit.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Amogh R B

Role: Team member (MSRIT_Waypoint)

Signature: *Amogh Rajeev Balla*

Date: 30 September 2026